

Forecasting FTSE 100 Monthly Returns: S&P 500, UK Bond Yields, and GBP/USD

Research Question

Do past movements in the S&P 500, UK 10-year bond yields, and the GBP/USD exchange rate help forecast FTSE 100 monthly returns?

Variable	Role	What it Measures
FTSE_RET	Dependent variable	FTSE 100 monthly return; UK stock market return
SP500_RET	Independent variable	S&P 500 monthly return; U.S. stock market movement
DUK10Y	Independent variable	Monthly change in UK 10-year government bond yield
GBPUSD_RET	Independent variable	Monthly change/return in GBP/USD exchange rate

Background & Part 1 Extension

- Part 1 found that S&P 500 quarterly returns strongly explained FTSE 100 quarterly returns
- UK 10-year bond-yield changes were not statistically significant in the part 1 model
- For this project, I extend the analysis by using monthly data instead of quarterly data
- I also add GBP/USD exchange-rate movements because the FTSE 100 includes many globally connected companies
- This project uses dynamic time-series tools such as VAR, Granger causality, impulse responses, and cointegration testing

Literature Review

- Stock Market Linkages, Bond Yields, and Exchange Rates
- **Efficient market theory** suggests that stock prices respond quickly to available information, which supports testing whether U.S. market movements are reflected in UK stock returns (Fama, 1970).
- **International stock markets are connected.** Eun and Shim find strong transmission effects among national stock markets, supporting the idea that S&P 500 movements may help explain or forecast FTSE 100 returns (Eun & Shim, 1989).
- **New York and London markets have direct linkages.** Hamao, Masulis, and Ng find short-run price and volatility connections across major global markets, including New York and London, which directly supports my U.S.–UK stock market comparison (Hamao, Masulis, & Ng, 1990).
- **Stock and bond markets can be connected through interest rates and economic news.** Campbell and Ammer show that stock and long-term bond returns are related to news about future returns, inflation, and real interest rates, supporting the use of UK 10-year bond-yield changes in my model (Campbell & Ammer, 1993).
- **Exchange rates may also matter for stock markets.** Ajayi and Mougoué find dynamic relationships between stock prices and exchange rates, supporting the addition of GBP/USD as a variable (Ajayi & Mougoué, 1996).

This project applies these ideas to recent UK monthly data and tests whether past movements help forecast FTSE 100 returns

Data Description

Data frequency: Monthly
Sample period: 2001M02 to 2025M12
Data sources: Investing.com
Main country focus: United Kingdom
Stock market studied: FTSE 100

- Why monthly data?
  - Provides more observations than quarterly data
  - Captures short-run financial-market movements
  - Better suited for VAR and Granger causality tests

Variables	Description	Transformation
FTSE	UK Stock Market Index	Monthly percent return: FTSE_RET
SP500	U.S. Stock Market Index	Monthly percent return: SP500_RET
UK10Y	UK 10-Year Bond Yield	Monthly change: DUK10Y
GBP_USD	GBP/USD Exchange Rate	Monthly percent return: GBPUSD_RET

Variables & Hypotheses

Variable	Role	Meaning
FTSE_RET	Dependent variable	FTSE 100 monthly return
SP500_RET	Independent variable	S&P 500 monthly return
DUK10Y	Independent variable	Monthly change in UK 10-year bond yield
GBPUSD_RET	Independent variable	Monthly return/change in GBP/USD

Hypotheses	H1: SP500_RET helps explain or forecast FTSE_RET	Expected sign: positive
	H2: DUK10Y helps explain or forecast FTSE_RET	Expected sign: negative or uncertain
	H3: GBPUSD_RET helps explain or forecast FTSE_RET	Expected sign: uncertain

Methodology

Methods used:	Graphs of level and transformed variables
	ADF unit root tests
	Benchmark OLS regression
	Interest-rate-only OLS regression
	VAR lag selection
	VAR(1) model
	Granger causality tests
	Impulse response functions
	Johansen cointegration test for VEC decision
Main model path:	Nonstationary levels → stationary returns/changes → OLS + VAR → Granger + IRF → Johansen/VEC decision

ADF Unit Root Test Results

The level variables are nonstationary, but the transformed return/change variables are stationary

Variable	ADF Statistic	p-value	Conclusion
FTSE	-0.179	0.938	Nonstationary
S&P 500	2.765	1.000	Nonstationary
UK10Y	-1.436	0.565	Nonstationary
GBP/USD	-1.346	0.609	Nonstationary
FTSE_RET	-17.531	0.000	Stationary
SP500_RET	-16.815	0.000	Stationary
DUK10Y	-17.497	0.000	Stationary
GBPUSD_RET	-16.342	0.000	Stationary

Benchmark OLS Results

Model

FTSE_RET = C + SP500_RET + DUK10Y + GBPUSD_RET

Variable	Coefficient	p-value	Result
SP500_RET	0.7459	0.0000	Significant positive effect
DUK10Y	0.7228	0.1946	Not statistically significant
GBPUSD_RET	-0.3155	0.0000	Significant negative effect

- R-squared = 0.6327
- Model explains about 63.3% of FTSE monthly return variation
- S&P 500 returns have the strongest relationship with FTSE returns
- UK 10-year bond-yield changes are not significant in the full model

Benchmark OLS Results

Dependent Variable: FTSE_RET
Method: Least Squares
Date: 05/29/26 Time: 14:52
Sample: 2001M03 2025M12
Included observations: 298

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-0.249049	0.136040	-1.830703	0.0682
SP500_RET	0.745915	0.033528	22.24746	0.0000
DUK10Y	0.722756	0.555961	1.300010	0.1946
GBPUSD_RET	-0.315472	0.059100	-5.337915	0.0000

R-squared	0.632673	Mean dependent var	0.247043
Adjusted R-squared	0.628925	S.D. dependent var	3.803109
S.E. of regression	2.316700	Akaike info criterion	4.531497
Sum squared resid	1577.927	Schwarz criterion	4.581122
Log likelihood	-671.1930	Hannan-Quinn criter.	4.551361
F-statistic	168.7921	Durbin-Watson stat	2.248837
Prob(F-statistic)	0.000000		

Interest-Rate-Only Regression

Dependent Variable: FTSE_RET
Method: Least Squares
Date: 05/29/26 Time: 14:58
Sample: 2001M03 2025M12
Included observations: 298

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.247766	0.220550	1.123401	0.2622
DUK10Y	0.544040	0.909711	0.598036	0.5503

R-squared	0.001207	Mean dependent var	0.247043
Adjusted R-squared	-0.002167	S.D. dependent var	3.803109
S.E. of regression	3.807228	Akaike info criterion	5.518369
Sum squared resid	4290.516	Schwarz criterion	5.543181
Log likelihood	-820.2369	Hannan-Quinn criter.	5.528301
F-statistic	0.357648	Durbin-Watson stat	2.038963
Prob(F-statistic)	0.550273		

- Model: FTSE_RET = C + DUK10Y

Variable	Coefficient	p-value	Result
DUK10Y	0.5440	0.5503	Not statistically significant

- Checked UK interest rates after dropping SP500_RET
- DUK10Y remains statistically insignificant
- R-squared = 0.0012, meaning the model explains almost none of the variation in FTSE returns

VAR Lag Length Selection

Criterion	Selected Lag
LR	1
FPE	1
AIC	1
SC	0
HQ	0

- Lag lengths from 0 to 4 months were tested
- LR, FPE, and AIC selected **1 lag**
- Since the project studies forecasting and dynamic relationships, I use **VAR(1)**
- VAR(1) allows last month's values to help explain this month's FTSE return
- **VAR result:** Past values show limited forecasting power; the U.S. market provides the strongest weak signal

VAR Lag Length Selection

VAR Lag Order Selection Criteria
Endogenous variables: FTSE_RET SP500_RET DUK10Y GBPUSD_RET
Exogenous variables: C
Date: 05/29/26 Time: 15:40
Sample: 2001M03 2025M12
Included observations: 294

Lag	LogL	LR	FPE	AIC	SC	HQ
0	-2171.162	NA	31.35926	14.79702	14.84713*	14.81709*
1	-2155.121	31.53490*	31.35108*	14.79674*	15.04733	14.89710
2	-2141.691	26.03784	31.90603	14.81423	15.26528	14.99486
3	-2134.869	13.04210	33.96666	14.87666	15.52817	15.13757
4	-2129.200	10.68188	36.45017	14.94694	15.79892	15.28813

* indicates lag order selected by the criterion
LR: sequential modified LR test statistic (each test at 5% level)
FPE: Final prediction error
AIC: Akaike information criterion
SC: Schwarz information criterion
HQ: Hannan-Quinn information criterion

Granger Causality Results

Test	p-value	Result
SP500_RET → FTSE_RET	0.0828	Weak evidence at 10%, not 5%
DUK10Y → FTSE_RET	0.4241	Not significant
GBPUSD_RET → FTSE_RET	0.8069	Not significant
All variables jointly → FTSE_RET	0.1667	Not significant at 5%

Key takeaway:

Past S&P 500 returns provide weak forecasting evidence, but past UK bond-yield changes and past GBP/USD returns do not significantly forecast FTSE returns

Granger Causality Results

VAR Granger Causality/Block Exogeneity Wald Tests
Date: 05/29/26 Time: 15:47
Sample: 2001M03 2025M12
Included observations: 297

Dependent variable: FTSE_RET

Excluded	Chi-sq	df	Prob.
SP500_RET	3.009072	1	0.0828
DUK10Y	0.638993	1	0.4241
GBPUSD_RET	0.059769	1	0.8069
All	5.070437	3	0.1667

Dependent variable: SP500_RET

Excluded	Chi-sq	df	Prob.
FTSE_RET	0.090130	1	0.7640
DUK10Y	0.020986	1	0.8848
GBPUSD_RET	0.565813	1	0.4519
All	0.575266	3	0.9021

Dependent variable: DUK10Y

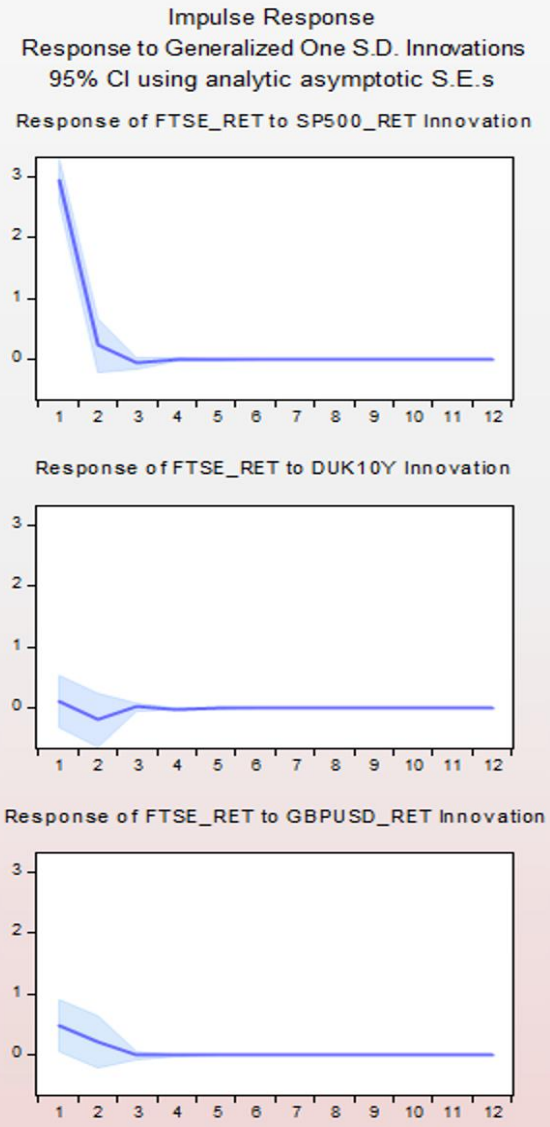
Excluded	Chi-sq	df	Prob.
FTSE_RET	1.189383	1	0.2755
SP500_RET	1.439913	1	0.2302
GBPUSD_RET	0.615601	1	0.4327
All	12.70574	3	0.0053

Dependent variable: GBPUSD_RET

Excluded	Chi-sq	df	Prob.
FTSE_RET	8.561082	1	0.0034
SP500_RET	4.346337	1	0.0371
DUK10Y	0.089688	1	0.7646

Impulse Response Results

- FTSE_RET responds most strongly to an S&P 500 return shock
- The S&P 500 shock effect is positive but fades after about 2–3 months
- UK bond-yield shocks have a very small effect on FTSE_RET
- GBP/USD shocks also have only a short-lived effect
- Overall, the dynamic effects are temporary, not persistent



Johansen Cointegration / VEC Decision

- ADF tests showed the level variables are nonstationary
- Johansen test checks whether nonstationary level variables share a long-run relationship
- Both tests indicate **no cointegration at the 5% level**
- Therefore, a VEC model is **not appropriate**
- The final dynamic model uses VAR with stationary return/change variables

Test	p-value for "No Cointegration"	Result
Trace Test	0.4111	No cointegration
Max-Eigenvalue Test	0.4392	No cointegration

Johansen Cointegration / VEC Decision

Summary

Johansen Cointegration Test

Date: 05/29/26 Time: 15:57
Sample: 2001M02 2025M12
Included observations: 299
Lags interval (in first differences): 1 to 1
Endogenous variables: FTSE SP500 UK10Y GBPUSD
Deterministic assumptions: Case 3 (Johansen-Hendry-Juselius): Cointegrating relationship includes a constant. Short-run dynamics include a constant.

Rank Tests

Unrestricted Cointegration Rank Test (Trace)

Hypothesized No. of CE(s)	Eigenvalue	Trace Statistic	0.05 Critical Value	Prob.** Critical Value
None	0.060976	35.70942	47.85613	0.4111
At most 1	0.036824	17.02403	29.79707	0.6383
At most 2	0.018691	5.880985	15.49471	0.7095
At most 3	0.000933	0.277317	3.841465	0.5985

Trace test indicates no cointegration at the 0.05 level
* denotes rejection of the hypothesis at the 0.05 level
**MacKinnon-Haug-Michelis (1999) p-values

Unrestricted Cointegration Rank Test (Max-eigenvalue)

Hypothesized No. of CE(s)	Eigenvalue	Max-Eigen Statistic	0.05 Critical Value	Prob.** Critical Value
None	0.060976	18.68539	27.58434	0.4392
At most 1	0.036824	11.14305	21.13162	0.6330
At most 2	0.018691	5.603668	14.26460	0.6644
At most 3	0.000933	0.277317	3.841465	0.5985

Max-eigenvalue test indicates no cointegration at the 0.05 level
* denotes rejection of the hypothesis at the 0.05 level
**MacKinnon-Haug-Michelis (1999) p-values

Unrestricted Coefficients

Unrestricted Cointegrating Coefficients (normalized by b'*S11*b=I):

FTSE	SP500	UK10Y	GBPUSD
-0.000510	0.001104	-0.395964	2.119989
-0.000116	0.000369	-0.501797	6.386514
-0.001543	0.000947	-0.083869	0.297668
0.000120	0.000416	0.516932	1.736194

Unrestricted Adjustment Coefficients (alpha):

D(FTSE)	26.14990	-0.912929	26.03281	1.458204
D(SP500)	18.05352	-11.02068	2.851485	1.827735
D(UK10Y)	0.025993	-0.004066	-0.000504	-0.006488
D(GBPUSD)	-0.001976	-0.006600	0.001389	-0.000144

Findings & Conclusion

The project finds strong same-month U.S.–UK stock market linkage, but limited forecasting power from past variables.

- FTSE 100 monthly returns are strongly related to same-month S&P 500 returns
- SP500_RET is positive and statistically significant in OLS
- GBPUSD_RET is significant and negative in OLS
- DUK10Y is not statistically significant in the full model or the interest-rate-only model
- Granger causality shows limited forecasting power
- Past S&P 500 returns provide weak evidence at 10%, but not 5%
- Past UK bond-yield and GBP/USD movements do not significantly forecast FTSE returns
- No cointegration was found, so VEC is not appropriate

Limitations

- The model shows statistical relationships, not guaranteed causation
- Only three explanatory variables are included
- Other variables may matter, such as:
  - Euro Stoxx 50 or DAX returns
  - UK inflation
  - Bank of England policy rate
  - Oil prices
  - Brexit, COVID-19, and 2008 crisis dummies
- Monthly data may still miss daily market reactions
- Future research could test additional variables or crisis-period effects

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